Curriculum Vitae

Mehedi Hasan
Research Assistant, Population Medicine and AMR Laboratory
Bangladesh Agricultural University, Bangladesh
Email: mehedi.1801112@bau.edu.bd, mhd7566@gmail.com
Contact: +8801838596615
LinkedIn: www.linkedin.com/in/mehedi-hasan-5b3404209
Portfolio: https://mehedix112.github.io/Portfolio_Mehedi-Hasan/
Orcid ID: 0009-0006-2571-7173

RESEARCH INTEREST

Life Science, Antimicrobial Resistance (AMR), One Health Epidemiology

EDUCATION

M.S. in Medicine **2024-2026**
CGPA: 3.696/4.00
Department of Medicine,
Bangladesh Agricultural University (BAU), Mymensingh

Master's thesis dissertation: Prevalence and antimicrobial resistance pattern of *Acinetobacter baumannii* from veterinary hospitals and pet animals in Bangladesh

Doctor of Veterinary Medicine (D.V.M.) **2018-2024**
CGPA: 3.577/4.00
Last 4 Semester CGPA: 3.671/4.00
Faculty of Veterinary Science,
Bangladesh Agricultural University, Mymensingh

TECHNICAL & STATISTICAL SKILLS

- **Molecular & Laboratory:** Bacterial isolation and culture, species-specific and multiplex PCR, antimicrobial susceptibility testing (disk diffusion/broth microdilution), microtiter plate biofilm assays, ELISA, total viable count (TVC), somatic cell count, protozoal count.
- **Statistical:** R and SPSS for mixed-effects and hierarchical modelling (GLMM, GEE, three-level hierarchical models), Linear, Logistic, Ordinal and Polynomial regression models, Monte Carlo Simulation, Multiple correspondence and Principal component analysis, DAG-based confounding analysis, and Kaplan-Meier survival analysis
- **Epidemiological & Spatial:** Outbreak investigation, QGIS-based spatial mapping, hypothesis testing, experimental study designs and risk-factor analysis.
- **Bioinformatics:** Phylogenetic reconstruction and amino acid substitution analysis using MEGA and BioEdit, Primer design

PUBLICATIONS

Published Articles

- Runa, M.A., Tasnuva, S., Rahman, A., Rahman, M.M., **Hasan, M.**, et al. (2026). Occurrence, virulence gene profile, and antimicrobial resistance of *Campylobacter* spp. in ready-to-eat and processed chicken and fish products in Mymensingh municipality of Bangladesh. International Journal of Food Science. https://doi.org/10.1155/ijfo/9923913

- **Hasan, M.**, Lira, S.A., Islam, A., et al (2026). Occurrence and antimicrobial resistance of *Acinetobacter baumannii* in cats, veterinary hospital environments and medical waste in Bangladesh. Veterinary Research Communications **(In press)**
- Rahman, A., **Hasan, M.**, Khatun, Z., et al. Epidemiology of recent outbreaks of Foot-and-mouth disease in cattle and emergence of SEA/Mya-98 topotype in Bangladesh. Bulletin of the National Research Center **(Accepted)**

Articles Under Review

- Lira, S.A., **Hasan, M.**, Islam, A., et al. Assessment of veterinary waste management practices and detection of antibiotic-resistant *Klebsiella pneumoniae* in veterinary wastes of Bangladesh. Under review at *One Health*
- Jahan, I., **Hasan, M.**, Sharmy, S.T., et al. Economic impact and epidemiology of Lumpy Skin Disease in cattle in Bangladesh. Under review at *Preventive Veterinary Medicine: Regional Studies and Reports*
- Miti, A.A., **Hasan, M.**, Suchy, S.A., et al. Influence of cow and herd level factors on somatic cell count and total bacterial count in dairy cow milk. Under review at *PlOS One*
- Moon. M.M., **Hasan, M.,** Keya, A.N., et al. Integrated assessment of disease prevention practices and antimicrobial stewardship risk in small-scale broiler farms in Bangladesh. Under review at *The Journal of Poultry Science*

Manuscripts in Preparation

- Impact evaluation of herd-health practices on metabolic disease
- Quantification of Antimicrobial and alternative to antimicrobial use in broiler farms in Bangladesh
- Factors associated with rumen health;
- Biofilm-forming *Salmonella* spp. and MRSA in poultry drinking water systems;
- Commensal *E. coli* carrying carbapenem resistance genes in retail chicken meat;
- Effect of egg yolk derived immunoglobulin Y (IgY) on calf diarrhoea

CONFERENCES / SEMINAR ABSTRACTS

Oral

- **Hasan, M.**, Lira, S.A., Islam, A., et al. (2026) First report of carbapenem-resistant *Acinetobacter baumannii* in veterinary hospitals and pet animals in Bangladesh 32nd BSVER ASCon, (Bangladesh Society for Veterinary Education and Research). **Presenting author**
- Lira, S.A., **Hasan, M.**, Islam, A., et al. (2026) Assessment of veterinary waste management practices and detection of antibiotic-resistant *Klebsiella pneumoniae* in veterinary wastes of Bangladesh. 2nd Bangladesh FETP Conference, 2026 (Field Epidemiology Training Program). **Presenting author**
- Ali, S.B., **Hasan, M.**, et al. (2026) Factors associated with rumen health parameters in crossbred dairy cattle under contrasting management systems in Bangladesh. 32nd BSVER ASCon. **Presenting author**
- Turna, A.A., Maliha, M., **Hasan, M.**, Rahman, A., et al. (2025) Isolation and characterization of biofilm-forming *Salmonella* spp. from poultry drinking water system. International Poultry Show and Seminar, World's Poultry Science Association, Bangladesh Branch (WPSA-BB)
- Parvin, M.S., Maliha, M., **Hasan, M.**, Rahman, A., et al. (2025) Exploring commensal *E. coli* carrying carbapenem resistance genes isolated from frozen and retail chicken meat in Bangladesh. International Poultry Show and Seminar, World's Poultry Science Association, Bangladesh Branch (WPSA-BB)

Poster

- Maliha, M., Turna, A.A., **Hasan, M.**, et al. (2025) Detection of biofilm-forming methicillin-resistant *Staphylococcus aureus* in drinking water systems of layer farms. World Veterinary Poultry Association Congress, Sarawak, Malaysia

Conferences/Seminar Attended: 2nd FETP Conference (2026), 32nd BSVER (2026), International Poultry Show and Seminar, WPSA-BB (2025); 30th BSVER Annual Scientific Conference (2024); CEVET BD Conference (2020)

RESEARCH EXPERIENCES

Research Assistant, Population Medicine and AMR Laboratory, BAU **2023-Present**
Principal Investigator: Prof. Dr. Md. Taohidul Islam
Contributed across multiple concurrent studies spanning antimicrobial resistance and quantification, surveillance, food safety, and veterinary and dairy epidemiology

Responsibilities

- Conducted extensive field-based sample collection from veterinary hospitals, pet animals, waste sites, commercial and household dairy and poultry farms across multiple districts
- Performed bacterial isolation, biochemical characterization, and molecular identification (species-specific and multiplex PCR) of key zoonotic and foodborne pathogens (*Acinetobacter baumannii*, *Klebsiella pneumoniae*, *Salmonella* spp., *S. aureus*, *L. monocytogenes*, *E. coli*, *Streptococcus* spp., and *Campylobacter* spp.)
- Contributed to the design and execution of a large-scale cross-sectional epidemiological and economic impact study on Lumpy Skin Disease (LSD) in cattle
- Antimicrobial use quantification through matrices like Used Daily Dose (UDD), Defined daily dose (DDD)
- Experimental designs and longitudinal cohort study implementation
- Bioinformatics study consisting of phylogenetic analysis, amino acid substitution, primer design etc.
- Conducted systematic literature reviews, data analysis, and contributing to manuscript preparation
- Mentoring 4 junior M.S. fellows and research assistants in laboratory methods, data analysis, scientific writing, and designing their MS thesis researches

PROFESSIONAL EXPERIENCES

Teaching Assistant, Department of Medicine, BAU **2024-2026**

- Conducted hands-on QGIS practical classes for first-semester Master's students, including live demonstration of spatial analysis workflows and direct supervision during sessions
- Assisted in course material development including lecture notes and examination question preparation, and contributed to student performance evaluation
- Provided informal one-on-one academic support to Master's students, assisting with research methodology, data analysis, and thesis-related queries outside scheduled classes

CLINICAL INTERNSHIP

Public and Private Hospitals, Clinics and Farm **2024**

Veterinary Hospitals and Clinics, Military Dairy Farms, BRAC Bull and Buck Station, Livestock Research Institute, Bangladesh National Zoo, Central Disease Investigation Laboratory (CDIL)

Activities

- Performed patient handling and restraint across multiple species including small animals, livestock, poultry, and birds, and assisted in clinical examination, history taking, and disease diagnosis alongside senior clinicians
- Participated in surgical procedures including spaying, neutering, Caesarean section, fracture management, amputation, hernia repair, and wound management as a surgical assistant
- Conducted post-mortem examinations of sheep, goat, calf, and chicken, with subsequent sample collection and processing for laboratory investigation

HONORS & AWARDS

- National Science and Technology (NST) Fellowship, Ministry of Science and Technology, Bangladesh

INTERNATIONAL EXPOSURE

- Participant (Offline), Sakura Science Exchange Program, Shimane University, Japan (2025) — trained in advanced microscopy, histology, and laboratory animal handling
- Participant, Hult Prize, Regional (South Asia) round (2024)

PROFESSIONAL ASSOCIATIONS

- Registered Veterinarian, Bangladesh Veterinary Council (Reg. No: 10722)
- Global Outreach - Student Membership, American Society for Microbiologist (ASM)

ACADEMIC & COMMUNITY SERVICE

- Member, Scientific and Publication Sub-committee, 32nd BSVER Annual Scientific Conference (2026)
- Student Representative, Social Harmony Committee, Mymensingh (2021–2024); Joint Secretary, Hashimukh voluntary organisation, BAU (2018–2022) — student governance and flood relief/public health outreach
- Conducted AMR awareness sessions for rural dairy farmers and supported livestock vaccination campaigns in Pabna and Mymensingh districts

LANGUAGE PROFICIENCY

- English – IELTS: 7.5 (Reading: 9, Listening: 7.5, Writing: 6.5, Speaking: 6.5)

REFERENCES

Prof. Dr. Md. Taohidul Islam
Dept. of Medicine, Bangladesh Agricultural University
Email: taohid@bau.edu.bd
Phone: +8801912910338

Prof. Dr. Mst. Sonia Parvin
Dept. of Medicine, Bangladesh Agricultural University
Email: soniaparvin@bau.edu.bd
Phone: +8801922471931